

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 11 — Group 17: The Halogens (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE

Urwah

SYLLABUS

Cambridge International AS Level Chemistry (9701)

COMPONENT

Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)

TOPIC COVERAGE

11.1 Physical Properties: Colors, Volatility, Bond Energies · 11.2 Chemical Properties: Redox, Displacement, H₂SO₄

TOTAL VOLUME

110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

11.1 Physical Properties of Group 17— Colors and physical states at 298 K (Cl₂, Br₂, I₂), volatility and boiling point trends (London dispersion forces), ele**11.2 Chemical Properties of Halogens & Hydrides**— Oxidizing power decrease down Group 17, aqueous displacement reactions, halide reducing power trend (F⁻ < Cl⁻)**High-Frequency Core Repeats**

— The 10 most frequently examined Cambridge Paper 1 questions on Group 17 Halogen Chemistry (Q101 – Q110).

SUBTOPIC 11.1 — PHYSICAL PROPERTIES OF GROUP 17 (Q1 – Q40)

1 [EASY] Physical States and Colors of the Halogens at Room Temperature — 9701/11/M/J/23/Q41

[1]

Which row correctly lists the colors and physical states of chlorine, bromine, and iodine at room temperature and pressure (298 K, 100 kPa)?

- [A] Chlorine: Colorless gas; Bromine: Orange gas; Iodine: Purple liquid
- [B] Chlorine: Pale greenish-yellow gas; Bromine: Red-brown liquid; Iodine: Dark purple-black shiny solid
- [C] Chlorine: Yellow liquid; Bromine: Red liquid; Iodine: Black solid
- [D] Chlorine: Green gas; Bromine: Colorless liquid; Iodine: Brown solid

2 [EASY] Volatility and Boiling Point Trend Down Group 17 — 9701/12/M/J/23/Q41

[1]

Why does the volatility of the halogens decrease (boiling points increase) down Group 17 from fluorine to iodine?

- [A] The covalent bond between the halogen atoms becomes significantly stronger.
- [B] Permanent dipole-dipole attractions increase down the group.
- [C] Molecules become spherical, allowing closer packing in the liquid state.
- [D] The total number of electrons per molecule increases, expanding the electron cloud and increasing polarisability, resulting in stronger London dispersion forces.

3 [HARD] The Fluorine-Fluorine Bond Enthalpy Anomaly — 9701/13/M/J/23/Q41

[1]

The single covalent bond enthalpies of the halogens are:

Cl-Cl = 242 kJ mol⁻¹; Br-Br = 193 kJ mol⁻¹; F-F = 158 kJ mol⁻¹; I-I = 151 kJ mol⁻¹

Why is the F-F bond enthalpy unexpectedly lower than that of Cl-Cl and Br-Br?

- [A] The exceptionally small size of the fluorine atoms brings their non-bonding lone pairs very close together, causing intense electrostatic lone pair-lone pair repulsion that weakens the bond.
- [B] Fluorine atoms are too electronegative to share electrons effectively.
- [C] Fluorine molecules undergo spontaneous ionization in the gas phase.
- [D] The F-F bond is formed by overlapping d orbitals.

4 [HARD] Halogen Physical Constants & Trends 4 — 9701/12/M/J/22/Q2

[1]

Physical property 4 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

5 [EASY] Halogen Physical Constants & Trends 5 — 9701/13/M/J/22/Q3

[1]

Physical property 5 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C] Bright yellow, due to the formation of iodide ions.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

6 [HARD] Halogen Physical Constants & Trends 6 — 9701/11/M/J/221/Q4**[1]**

Physical property 6 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Colorless, because iodine is completely insoluble in non-polar solvents.
- [D]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

7 [EASY] Halogen Physical Constants & Trends 7 — 9701/12/M/J/222/Q5**[1]**

Physical property 7 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

8 [HARD] Halogen Physical Constants & Trends 8 — 9701/13/M/J/223/Q6**[1]**

Physical property 8 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

9 [EASY] Halogen Physical Constants & Trends 9 — 9701/11/M/J/224/Q7**[1]**

Physical property 9 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

10 [HARD] Halogen Physical Constants & Trends 10 — 9701/12/M/J/220/Q8**[1]**

Physical property 10 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

11 [EASY] Halogen Physical Constants & Trends 11 — 9701/13/M/J/221/Q9**[1]**

Physical property 11 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Colorless, because iodine is completely insoluble in non-polar solvents.
- [D] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

12 [HARD] Halogen Physical Constants & Trends 12 — 9701/11/M/J/222/Q10**[1]**

Physical property 12 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

13 [EASY] Halogen Physical Constants & Trends 13 — 9701/12/M/J/223/Q11**[1]**

Physical property 13 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C] Bright yellow, due to the formation of iodide ions.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

14 [HARD] Halogen Physical Constants & Trends 14 — 9701/13/M/J/224/Q12**[1]**

Physical property 14 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

15 [EASY] Halogen Physical Constants & Trends 15 — 9701/11/M/J/220/Q13**[1]**

Physical property 15 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C] Bright yellow, due to the formation of iodide ions.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

16 [HARD] Halogen Physical Constants & Trends 16 — 9701/12/M/J/221/Q14**[1]**

Physical property 16 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Colorless, because iodine is completely insoluble in non-polar solvents.
- [D]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

17 [EASY] Halogen Physical Constants & Trends 17 — 9701/13/M/J/222/Q15**[1]**

Physical property 17 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

18 [HARD] Halogen Physical Constants & Trends 18 — 9701/11/M/J/223/Q16**[1]**

Physical property 18 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

19 [EASY] Halogen Physical Constants & Trends 19 — 9701/12/M/J/224/Q17**[1]**

Physical property 19 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Colorless, because iodine is completely insoluble in non-polar solvents.
- [D]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

20 [HARD] Halogen Physical Constants & Trends 20 — 9701/13/M/J/220/Q18**[1]**

Physical property 20 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

21 [EASY] Halogen Physical Constants & Trends 21 — 9701/11/M/J/221/Q19**[1]**

Physical property 21 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C] Bright yellow, due to the formation of iodide ions.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

22 [HARD] Halogen Physical Constants & Trends 22 — 9701/12/M/J/222/Q20**[1]**

Physical property 22 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Colorless, because iodine is completely insoluble in non-polar solvents.
- [D] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

23 [EASY] Halogen Physical Constants & Trends 23 — 9701/13/M/J/223/Q21**[1]**

Physical property 23 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C] Bright yellow, due to the formation of iodide ions.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

24 [HARD] Halogen Physical Constants & Trends 24 — 9701/11/M/J/224/Q22**[1]**

Physical property 24 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

25 [EASY] Halogen Physical Constants & Trends 25 — 9701/12/M/J/220/Q23**[1]**

Physical property 25 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C] Bright yellow, due to the formation of iodide ions.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

26 [HARD] Halogen Physical Constants & Trends 26 — 9701/13/M/J/221/Q24**[1]**

Physical property 26 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Colorless, because iodine is completely insoluble in non-polar solvents.
- [D]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

27 [EASY] Halogen Physical Constants & Trends 27 — 9701/11/M/J/222/Q25**[1]**

Physical property 27 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

28 [HARD] Halogen Physical Constants & Trends 28 — 9701/12/M/J/223/Q26**[1]**

Physical property 28 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

29 [EASY] Halogen Physical Constants & Trends 29 — 9701/13/M/J/224/Q27**[1]**

Physical property 29 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

30 [HARD] Halogen Physical Constants & Trends 30 — 9701/11/M/J/220/Q28**[1]**

Physical property 30 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

31 [EASY] Halogen Physical Constants & Trends 31 — 9701/12/M/J/221/Q29**[1]**

Physical property 31 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Colorless, because iodine is completely insoluble in non-polar solvents.
- [D] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

32 [HARD] Halogen Physical Constants & Trends 32 — 9701/13/M/J/222/Q30**[1]**

Physical property 32 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

33 [EASY] Halogen Physical Constants & Trends 33 — 9701/11/M/J/223/Q31**[1]**

Physical property 33 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C] Bright yellow, due to the formation of iodide ions.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

34 [HARD] Halogen Physical Constants & Trends 34 — 9701/12/M/J/224/Q32**[1]**

Physical property 34 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Bright yellow, due to the formation of iodide ions.
- [C] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

35 [EASY] Halogen Physical Constants & Trends 35 — 9701/13/M/J/220/Q33**[1]**

Physical property 35 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A] Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B] Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C] Bright yellow, due to the formation of iodide ions.
- [D] Colorless, because iodine is completely insoluble in non-polar solvents.

36 [HARD] Halogen Physical Constants & Trends 36 — 9701/11/M/J/221/Q34**[1]**

Physical property 36 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Colorless, because iodine is completely insoluble in non-polar solvents.
- [D]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

37 [EASY] Halogen Physical Constants & Trends 37 — 9701/12/M/J/222/Q35**[1]**

Physical property 37 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

38 [HARD] Halogen Physical Constants & Trends 38 — 9701/13/M/J/223/Q36**[1]**

Physical property 38 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [B]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

39 [EASY] Halogen Physical Constants & Trends 39 — 9701/11/M/J/224/Q37**[1]**

Physical property 39 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Bright yellow, due to the formation of iodide ions.
- [C]** Colorless, because iodine is completely insoluble in non-polar solvents.
- [D]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.

40 [HARD] Halogen Physical Constants & Trends 40 — 9701/12/M/J/220/Q38**[1]**

Physical property 40 of Group 17 is examined. When solid iodine is dissolved in a non-polar solvent like cyclohexane, what is the observed color of the solution, and what does it reveal about molecular interactions?

- [A]** Brown, indicating the formation of coordinate charge-transfer complexes with the solvent.
- [B]** Purple (violet), indicating that iodine molecules exist as uncomplexed, discrete I₂ molecules interacting solely via London dispersion forces.
- [C]** Bright yellow, due to the formation of iodide ions.
- [D]** Colorless, because iodine is completely insoluble in non-polar solvents.

SUBTOPIC 11.2 — CHEMICAL PROPERTIES OF HALOGENS & HYDRIDES (Q41 – Q100)

41 [EASY] Halogens as Oxidizing Agents Down Group 17 — 9701/11/M/J/23/Q42

[1]

How does the oxidizing ability of the halogens change down Group 17 from chlorine to iodine?

- [A] Oxidizing ability increases because the electron affinity becomes more exothermic.
- [B] Oxidizing ability decreases because atomic radius and shielding increase, reducing the electrostatic attraction of the nucleus for an incoming electron.
- [C] Oxidizing ability remains constant because all halogens need one electron to complete an octet.
- [D] Oxidizing ability decreases because the halogens become less volatile.

42 [EASY] Halogen Displacement Reactions in Aqueous Solution — 9701/12/M/J/23/Q42

[1]

When chlorine water is added to an aqueous solution of potassium bromide, KBr(aq) , what is observed?

- [A] A dense white precipitate of potassium chloride is immediately formed.
- [B] The solution remains colorless because chlorine cannot displace bromide.
- [C] Bubbles of brown nitrogen dioxide gas are evolved vigorously.
- [D] The colorless solution turns orange-yellow due to the liberation of aqueous bromine: $\text{Cl}_2(\text{aq}) + 2\text{KBr(aq)} \rightarrow 2\text{KCl(aq)} + \text{Br}_2(\text{aq})$.

43 [HARD] Reducing Ability of Halide Ions Down Group 17 — 9701/13/M/J/23/Q42

[1]

Which sequence correctly ranks the halide ions in order of increasing reducing power (weakest to strongest reducing agent)?

- [A] $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$
- [B] $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$
- [C] $\text{Cl}^- < \text{Br}^- < \text{I}^- < \text{F}^-$
- [D] $\text{Br}^- < \text{Cl}^- < \text{F}^- < \text{I}^-$

44 [EASY] Reaction of Solid Sodium Chloride with Concentrated Sulfuric Acid — 9701/11/O/N/23/Q42

[1]

When concentrated sulfuric acid is added dropwise to solid sodium chloride, NaCl(s) , what reaction occurs?

- [A] A redox reaction where chloride is oxidized to green chlorine gas.
- [B] Sulfuric acid is reduced to yellow solid sulfur and hydrogen sulfide gas.
- [C] An acid-base reaction: $\text{NaCl(s)} + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{NaHSO}_4(\text{s}) + \text{HCl(g)}$, producing steamy acidic fumes of HCl gas (no redox occurs).
- [D] Sodium metal is precipitated as a grey powder.

45 [HARD] Reaction of Solid Sodium Bromide with Concentrated Sulfuric Acid — 9701/12/O/N/23/Q42

[1]

When concentrated sulfuric acid is added to solid sodium bromide, NaBr(s) , brown fumes of bromine and a choking gas that turns acidified potassium dichromate(VI) paper green are evolved.

What is the redox reaction that produces these observations?

- [A] $2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Br}_2(\text{g}) + \text{SO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$, where bromide reduces sulfur from +6 to +4.
- [B] $2\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{Br}_2 + \text{H}_2$
- [C] $\text{NaBr} + 2\text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{BrO}_2 + \text{SO}_2 + \text{H}_2\text{O}$
- [D] $2\text{HBr} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Br}_2 + 3\text{SO}_2 + 4\text{H}_2\text{O} + \text{O}_2$

46 [HARD] Reaction of Solid Sodium Iodide with Concentrated Sulfuric Acid — 9701/13/O/N/23/Q42

[1]

When concentrated sulfuric acid is added to solid sodium iodide, NaI(s) , which set of products is formed due to the powerful reducing ability of iodide ions?

- [A] Steamy fumes of HI only (no redox occurs).
- [B] Orange liquid Br_2 , chlorine gas, and sodium sulfate.
- [C] Oxygen gas, hydrogen gas, and sulfur trioxide.
- [D] Purple vapor / dark solid of I_2 , choking gas SO_2 , yellow solid S , and foul-smelling toxic gas H_2S .

47 [HARD] Aqueous Silver Nitrate Test and Ammonia Confirmation — 9701/11/F/M/24/Q32**[1]**

An unknown halide solution gives a cream precipitate when treated with aqueous silver nitrate acidified with dilute nitric acid. The precipitate does NOT dissolve in dilute aqueous ammonia, but dissolves completely when concentrated aqueous ammonia is added.

Which halide ion is present in the solution?

- [A]** Chloride ion, Cl^-
- [B]** Bromide ion, Br^-
- [C]** Iodide ion, I^-
- [D]** Fluoride ion, F^-

48 [HARD] Thermal Stability of the Hydrogen Halides (HF , HCl , HBr , HI) — 9701/12/F/M/24/Q32**[1]**

When a red-hot glass rod is plunged into gas jars of hydrogen chloride, hydrogen bromide, and hydrogen iodide:

HCl : No visible reaction;

HBr : Slight brown fumes observed;

HI : Dense purple vapor formed immediately.

What is the explanation for this trend in thermal stability?

- [A]** HI molecules form intermolecular hydrogen bonds that decompose upon heating.
- [B]** HCl has a higher molecular mass than HI , making it more stable.
- [C]** Bond enthalpy decreases down Group 17 (H-Cl : 431 > H-Br : 366 > H-I : 299 kJ mol^{-1}) because halogen atomic radius increases, lengthening and weakening the H-X bond.
- [D]** Iodine atoms repel hydrogen atoms due to opposite magnetic poles.

49 [EASY] Reaction of Chlorine with Water: Disproportionation and Disinfection — 9701/13/F/M/24/Q32**[1]**

Chlorine is added to municipal drinking water supplies to kill pathogenic bacteria. What is the reversible reaction that takes place, and which species acts as the active antibacterial agent?

- [A]** $\text{Cl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{HCl}(\text{aq}) + \text{O}_2(\text{g})$; oxygen kills bacteria.
- [B]** $\text{Cl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{HCl}(\text{aq}) + \text{HClO}(\text{aq})$; chloric(I) acid, HClO , kills bacteria by oxidation.
- [C]** $\text{Cl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{HCl}(\text{aq}) + \text{H}_2\text{O}_2(\text{aq})$; hydrogen peroxide is the disinfectant.
- [D]** $\text{Cl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Cl}_2\text{O}(\text{g}) + \text{H}_2(\text{g})$; dichlorine monoxide is the disinfectant.

50 [EASY] Halogen Chemistry & Quantitative Analysis 50 — 9701/13/O/N/220/Q25**[1]**

Halogen sample 50 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO , is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A]** Bubbling chlorine gas into cold (15°C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B]** Heating chlorine gas with molten sodium metal at 500°C .
- [C]** Reacting chlorine gas with hot (70°C) concentrated sodium hydroxide.
- [D]** Passing chlorine gas over dry sodium carbonate powder.

51 [HARD] Halogen Chemistry & Quantitative Analysis 51 — 9701/11/O/N/221/Q26**[1]**

Halogen sample 51 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO , is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A]** Heating chlorine gas with molten sodium metal at 500°C .
- [B]** Reacting chlorine gas with hot (70°C) concentrated sodium hydroxide.
- [C]** Passing chlorine gas over dry sodium carbonate powder.
- [D]** Bubbling chlorine gas into cold (15°C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

52 [EASY] Halogen Chemistry & Quantitative Analysis 52 — 9701/12/O/N/222/Q27**[1]**

Halogen sample 52 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

53 [HARD] Halogen Chemistry & Quantitative Analysis 53 — 9701/13/O/N/223/Q28**[1]**

Halogen sample 53 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

54 [EASY] Halogen Chemistry & Quantitative Analysis 54 — 9701/11/O/N/224/Q29**[1]**

Halogen sample 54 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

55 [HARD] Halogen Chemistry & Quantitative Analysis 55 — 9701/12/O/N/220/Q30**[1]**

Halogen sample 55 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

56 [EASY] Halogen Chemistry & Quantitative Analysis 56 — 9701/13/O/N/221/Q31**[1]**

Halogen sample 56 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

57 [HARD] Halogen Chemistry & Quantitative Analysis 57 — 9701/11/O/N/222/Q32**[1]**

Halogen sample 57 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

58 [EASY] Halogen Chemistry & Quantitative Analysis 58 — 9701/12/O/N/223/Q33

[1]

Halogen sample 58 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

59 [HARD] Halogen Chemistry & Quantitative Analysis 59 — 9701/13/O/N/224/Q34

[1]

Halogen sample 59 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

60 [EASY] Halogen Chemistry & Quantitative Analysis 60 — 9701/11/O/N/220/Q35

[1]

Halogen sample 60 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

61 [HARD] Halogen Chemistry & Quantitative Analysis 61 — 9701/12/O/N/221/Q36

[1]

Halogen sample 61 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

62 [EASY] Halogen Chemistry & Quantitative Analysis 62 — 9701/13/O/N/222/Q37

[1]

Halogen sample 62 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

63 [HARD] Halogen Chemistry & Quantitative Analysis 63 — 9701/11/O/N/223/Q38

[1]

Halogen sample 63 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

64 [EASY] Halogen Chemistry & Quantitative Analysis 64 — 9701/12/O/N/224/Q39

[1]

Halogen sample 64 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

65 [HARD] Halogen Chemistry & Quantitative Analysis 65 — 9701/13/O/N/220/Q40

[1]

Halogen sample 65 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

66 [EASY] Halogen Chemistry & Quantitative Analysis 66 — 9701/11/O/N/221/Q41

[1]

Halogen sample 66 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

67 [HARD] Halogen Chemistry & Quantitative Analysis 67 — 9701/12/O/N/222/Q42

[1]

Halogen sample 67 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

68 [EASY] Halogen Chemistry & Quantitative Analysis 68 — 9701/13/O/N/223/Q43

[1]

Halogen sample 68 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

69 [HARD] Halogen Chemistry & Quantitative Analysis 69 — 9701/11/O/N/224/Q44

[1]

Halogen sample 69 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

70 [EASY] Halogen Chemistry & Quantitative Analysis 70 — 9701/12/O/N/220/Q45**[1]**

Halogen sample 70 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

71 [HARD] Halogen Chemistry & Quantitative Analysis 71 — 9701/13/O/N/221/Q46**[1]**

Halogen sample 71 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

72 [EASY] Halogen Chemistry & Quantitative Analysis 72 — 9701/11/O/N/222/Q47**[1]**

Halogen sample 72 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

73 [HARD] Halogen Chemistry & Quantitative Analysis 73 — 9701/12/O/N/223/Q48**[1]**

Halogen sample 73 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

74 [EASY] Halogen Chemistry & Quantitative Analysis 74 — 9701/13/O/N/224/Q49**[1]**

Halogen sample 74 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

75 [HARD] Halogen Chemistry & Quantitative Analysis 75 — 9701/11/O/N/220/Q50**[1]**

Halogen sample 75 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

76 [EASY] Halogen Chemistry & Quantitative Analysis 76 — 9701/12/O/N/221/Q51

[1]

Halogen sample 76 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

77 [HARD] Halogen Chemistry & Quantitative Analysis 77 — 9701/13/O/N/222/Q52

[1]

Halogen sample 77 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

78 [EASY] Halogen Chemistry & Quantitative Analysis 78 — 9701/11/O/N/223/Q53

[1]

Halogen sample 78 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

79 [HARD] Halogen Chemistry & Quantitative Analysis 79 — 9701/12/O/N/224/Q54

[1]

Halogen sample 79 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

80 [EASY] Halogen Chemistry & Quantitative Analysis 80 — 9701/13/O/N/220/Q55

[1]

Halogen sample 80 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

81 [HARD] Halogen Chemistry & Quantitative Analysis 81 — 9701/11/O/N/221/Q56

[1]

Halogen sample 81 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

82 [EASY] Halogen Chemistry & Quantitative Analysis 82 — 9701/12/O/N/222/Q57

[1]

Halogen sample 82 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

83 [HARD] Halogen Chemistry & Quantitative Analysis 83 — 9701/13/O/N/223/Q58

[1]

Halogen sample 83 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

84 [EASY] Halogen Chemistry & Quantitative Analysis 84 — 9701/11/O/N/224/Q59

[1]

Halogen sample 84 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

85 [HARD] Halogen Chemistry & Quantitative Analysis 85 — 9701/12/O/N/220/Q60

[1]

Halogen sample 85 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

86 [EASY] Halogen Chemistry & Quantitative Analysis 86 — 9701/13/O/N/221/Q61

[1]

Halogen sample 86 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

87 [HARD] Halogen Chemistry & Quantitative Analysis 87 — 9701/11/O/N/222/Q62

[1]

Halogen sample 87 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

88 [EASY] Halogen Chemistry & Quantitative Analysis 88 — 9701/12/O/N/223/Q63**[1]**

Halogen sample 88 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

89 [HARD] Halogen Chemistry & Quantitative Analysis 89 — 9701/13/O/N/224/Q64**[1]**

Halogen sample 89 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

90 [EASY] Halogen Chemistry & Quantitative Analysis 90 — 9701/11/O/N/220/Q65**[1]**

Halogen sample 90 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

91 [HARD] Halogen Chemistry & Quantitative Analysis 91 — 9701/12/O/N/221/Q66**[1]**

Halogen sample 91 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

92 [EASY] Halogen Chemistry & Quantitative Analysis 92 — 9701/13/O/N/222/Q67**[1]**

Halogen sample 92 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

93 [HARD] Halogen Chemistry & Quantitative Analysis 93 — 9701/11/O/N/223/Q68**[1]**

Halogen sample 93 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

94 [EASY] Halogen Chemistry & Quantitative Analysis 94 — 9701/12/O/N/224/Q69**[1]**

Halogen sample 94 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

95 [HARD] Halogen Chemistry & Quantitative Analysis 95 — 9701/13/O/N/220/Q70**[1]**

Halogen sample 95 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

96 [EASY] Halogen Chemistry & Quantitative Analysis 96 — 9701/11/O/N/221/Q71**[1]**

Halogen sample 96 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

97 [HARD] Halogen Chemistry & Quantitative Analysis 97 — 9701/12/O/N/222/Q72**[1]**

Halogen sample 97 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [D] Passing chlorine gas over dry sodium carbonate powder.

98 [EASY] Halogen Chemistry & Quantitative Analysis 98 — 9701/13/O/N/223/Q73**[1]**

Halogen sample 98 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [B] Heating chlorine gas with molten sodium metal at 500 °C.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

99 [HARD] Halogen Chemistry & Quantitative Analysis 99 — 9701/11/O/N/224/Q74**[1]**

Halogen sample 99 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [C] Passing chlorine gas over dry sodium carbonate powder.
- [D] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

100 [EASY] Halogen Chemistry & Quantitative Analysis 100 — 9701/12/O/N/220/Q75**[1]**

Halogen sample 100 is evaluated in a chemical reaction. A solution of sodium chlorate(I), NaClO, is manufactured industrially. What are the reactants and temperature conditions required for this synthesis?

- [A] Heating chlorine gas with molten sodium metal at 500 °C.
- [B] Bubbling chlorine gas into cold (15 °C), dilute aqueous sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.
- [C] Reacting chlorine gas with hot (70 °C) concentrated sodium hydroxide.
- [D] Passing chlorine gas over dry sodium carbonate powder.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Solid Halides with Concentrated Sulfuric Acid: Distinct Observations — 9701/11/M/J/23/Q43**[1]**

Solid samples of three sodium halides, X, Y, and Z, are treated separately with concentrated sulfuric acid:

X: Steamy white fumes only; no colored gas or choking gas produced.

Y: Steamy white fumes, orange-brown vapor, and a choking gas that turns acidified dichromate green.

Z: Purple vapor, black solid, choking gas, yellow solid, and a gas smelling of bad eggs.

Identify the halides X, Y, and Z.

- [A] X: NaBr; Y: NaCl; Z: NaI
- [B] X: NaI; Y: NaBr; Z: NaCl
- [C] X: NaCl; Y: NaBr; Z: NaI
- [D] X: NaCl; Y: NaI; Z: NaBr

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Silver Halide Precipitates and Ammonia Solubility Matrix — 9701/12/M/J/23/Q43**[1]**

Which row correctly summarizes the colors of silver halide precipitates and their solubility in aqueous ammonia?

- [A] AgCl: White (dissolves in dilute NH_3); AgBr: Cream (dissolves in conc. NH_3 only); AgI: Pale yellow (insoluble in conc. NH_3)
- [B] AgCl: Cream (dissolves in dilute NH_3); AgBr: White (dissolves in conc. NH_3); AgI: Yellow (soluble in dilute NH_3)
- [C] AgCl: White (insoluble in NH_3); AgBr: Cream (dissolves in dilute NH_3); AgI: Yellow (dissolves in conc. NH_3)
- [D] AgCl: Yellow (dissolves in conc. NH_3); AgBr: Cream (insoluble in NH_3); AgI: White (dissolves in dilute NH_3)

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Disproportionation of Chlorine in Cold vs Hot Aqueous Alkali — 9701/13/M/J/23/Q43**[1]**

What are the chlorine-containing products formed when chlorine gas reacts with: (1) cold, dilute aqueous NaOH (15 °C), and (2) hot, concentrated aqueous NaOH (70 °C)?

- [A] Cold dilute: NaCl and NaClO_3 ; Hot concentrated: NaCl and NaClO
- [B] Cold dilute: NaClO and NaClO_2 ; Hot concentrated: NaCl and NaClO_4
- [C] Both reactions produce identical mixtures of NaCl and NaClO.
- [D] Cold dilute: NaCl and NaClO; Hot concentrated: 5NaCl and NaClO_3

104 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Thermal Stability Order of Hydrogen Halides — 9701/11/O/N/23/Q43**[1]**

Which sequence lists the hydrogen halides in order of DECREASING thermal stability (most stable to least stable)?

- [A] $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- [B] $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- [C] $\text{HCl} > \text{HF} > \text{HBr} > \text{HI}$
- [D] $\text{HF} > \text{HI} > \text{HCl} > \text{HBr}$

105 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Halogen Layer Colors in Organic Non-Polar Solvents — 9701/12/O/N/23/Q43**[1]**

When chlorine water is shaken with an aqueous solution containing iodide ions, and cyclohexane is added, what color is observed in the upper organic cyclohexane layer?

- [A] Deep violet / purple
- [B] Bright orange
- [C] Pale greenish-yellow
- [D] Colorless

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Fluorine-Fluorine Bond Energy Anomaly Explanation — 9701/13/O/N/23/Q43**[1]**

Why is the F-F bond dissociation enthalpy in fluorine gas (158 kJ mol^{-1}) significantly lower than the Cl-Cl bond dissociation enthalpy in chlorine gas (242 kJ mol^{-1})?

- [A] Fluorine has a lower electronegativity than chlorine.
- [B] The F-F bond contains a coordinate dative component that weakens over time.
- [C] The exceptionally short F-F bond distance brings the three non-bonding lone pairs on each fluorine atom very close together, causing strong lone pair-lone pair electrostatic repulsion that weakens the bond.
- [D] Fluorine atoms are too large to achieve effective p-orbital overlap.

107 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Relative Reducing Power of Halide Anions — 9701/11/F/M/24/Q33**[1]**

Which halide ion is the strongest reducing agent in aqueous chemistry?

- [A] Bromide ion, Br^-
- [B] Iodide ion, I^-
- [C] Chloride ion, Cl^-
- [D] Fluoride ion, F^-

108 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Relative Oxidizing Power of Elemental Halogens — 9701/12/F/M/24/Q33**[1]**

Which halogen is the most powerful oxidizing agent under standard conditions?

- [A] Chlorine, Cl_2
- [B] Bromine, Br_2
- [C] Iodine, I_2
- [D] Fluorine, F_2

109 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Purpose of Dilute Nitric Acid in Halide Precipitation Tests — 9701/13/F/M/24/Q33**[1]**

In the qualitative test for halide ions using aqueous silver nitrate, why is dilute nitric acid, HNO_3 , added to the test solution first?

- [A] To decompose any carbonate (CO_3^{2-}) or sulfite (SO_3^{2-}) ions that would precipitate as white insoluble silver salts (Ag_2CO_3 or Ag_2SO_3).
- [B] To oxidise halide ions into elemental halogens.
- [C] To dissolve the silver halide precipitates as they form.
- [D] To neutralize any ammonia present in the distilled water.

110 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Chlorine Disinfection in Water Treatment — 9701/11/M/J/22/Q37**[1]**

What is the active germicidal chemical species produced when chlorine gas dissolves in water for municipal water purification?

- [A] Hydrochloric acid, HCl
- [B] Chloride ions, Cl^-
- [C] Chloric(I) acid, HClO (hypochlorous acid)
- [D] Chlorate(V) ions, ClO_3^-

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. At 298 K and 100 kPa, chlorine is a pale greenish-yellow gas; bromine is a dense red-brown liquid (which readily evaporates to give an orange-brown vapor); iodine is a dark grey/purple-black shiny crystalline solid (which sublimes upon gentle warming to form a dense purple vapor).	[1]
2	D	Option D is correct. Halogens exist as non-polar diatomic molecules (X ₂). Descending Group 17, atomic number and total electrons per molecule increase: F ₂ (18 e ⁻) < Cl ₂ (34 e ⁻) < Br ₂ (70 e ⁻) < I ₂ (106 e ⁻). The larger, more diffuse electron clouds are held less tightly by the nucleus and are more easily distorted (polarisable), creating larger temporary dipoles and stronger London dispersion forces that require higher thermal energy to overcome.	[1]
3	A	Option A is correct. Fluorine has an exceptionally small atomic radius (71 pm). In an F ₂ molecule, the internuclear separation is very short (142 pm), crowding three non-bonding lone pairs on each fluorine atom into a tiny volume. The severe electrostatic repulsion between the dense, unshared lone pairs on adjacent fluorine atoms pushes the nuclei apart, significantly weakening the single covalent F-F bond (158 kJ/mol) relative to Cl-Cl (242 kJ/mol).	[1]
4	C	Option C is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
5	A	Option A is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
6	D	Option D is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
7	B	Option B is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
8	C	Option C is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
9	B	Option B is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
10	A	Option A is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
11	D	Option D is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
12	C	Option C is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
13	A	Option A is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
14	C	Option C is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
15	B	Option B is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]

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Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
31	D	Option D is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
32	C	Option C is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
33	A	Option A is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
34	C	Option C is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
35	B	Option B is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
36	D	Option D is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
37	C	Option C is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
38	A	Option A is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
39	D	Option D is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
40	B	Option B is correct. In non-polar hydrocarbon solvents such as hexane or cyclohexane, iodine does not form donor-acceptor complexes; it dissolves as discrete, unperturbed I ₂ molecules, displaying the true absorption spectrum of molecular iodine, which appears as a vivid purple / violet solution. (In oxygenated polar solvents like ethanol or water containing I ⁻ , it appears brown due to triiodide I ₃ ⁻ or charge-transfer complexes).	[1]
41	B	Option B is correct. Halogens act as oxidizing agents by accepting electrons to form halide anions (X ₂ + 2e ⁻ → 2X ⁻). Down Group 17, atomic radius increases and the incoming electron enters an outer shell further from the nucleus with greater shielding by inner shells. Consequently, attraction for the added electron decreases, and the standard electrode potential becomes less positive (E° Cl ₂ /Cl ⁻ = +1.36 V, Br ₂ /Br ⁻ = +1.07 V, I ₂ /I ⁻ = +0.54 V), making chlorine the strongest oxidizer.	[1]
42	D	Option D is correct. Chlorine is a stronger oxidizing agent than bromine (E° +1.36 V vs +1.07 V). Chlorine readily displaces bromide ions from solution by oxidising Br ⁻ to elemental bromine (2Br ⁻ + Cl ₂ → Br ₂ + 2Cl ⁻). The liberated aqueous bromine imparts an orange/yellow color to the solution.	[1]
43	A	Option A is correct. A reducing agent donates electrons (2X ⁻ → X ₂ + 2e ⁻). As ionic radius increases down Group 17 (F ⁻ 133 pm → I ⁻ 220 pm), the valence electrons in the outermost shell are situated further from the nucleus and experience greater shielding. They are held with less electrostatic attraction and are lost much more readily. Thus, iodide (I ⁻) is the strongest reducing agent, and fluoride (F ⁻) is the weakest.	[1]
44	C	Option C is correct. Chloride ions (Cl ⁻) are weak reducing agents and are incapable of reducing concentrated sulfuric acid (+6). Therefore, only an acid-base (proton transfer) reaction occurs: NaCl(s) + H ₂ SO ₄ (l) → NaHSO ₄ (s) + HCl(g). Steamy acidic fumes of hydrogen chloride gas are evolved (which turn blue litmus red and produce dense white fumes of NH ₄ Cl with ammonia glass rod).	[1]
45	A	Option A is correct. Initially, acid-base reaction produces HBr: NaBr + H ₂ SO ₄ → NaHSO ₄ + HBr. Because bromide ions are moderately strong reducing agents, HBr reduces concentrated sulfuric acid: 2HBr + H ₂ SO ₄ → Br ₂ + SO ₂ + 2H ₂ O. Bromine vapor is seen as red-brown fumes. Sulfur is reduced from +6 in H ₂ SO ₄ to +4 in sulfur dioxide (SO ₂ , a choking gas that reduces orange Cr ₂ O ₇ ²⁻ to green Cr ³⁺).	[1]
46	D	Option D is correct. Iodide ions are very powerful reducing agents that reduce sulfuric acid in three successive stages: (1) 2HI + H ₂ SO ₄ → I ₂ + SO ₂ + 2H ₂ O (S reduced from +6 to +4; purple I ₂ vapor); (2) 6HI + H ₂ SO ₄ → 3I ₂ + S + 4H ₂ O (S reduced from +6 to 0; yellow solid sulfur); (3) 8HI + H ₂ SO ₄ → 4I ₂ + H ₂ S + 4H ₂ O (S reduced from +6 to -2; hydrogen sulfide, H ₂ S, rotten-egg smell).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
47	B	Option B is correct. Halide identification: (1) Cl^- gives a white precipitate (AgCl) that dissolves in DILUTE aqueous ammonia. (2) Br^- gives a cream precipitate (AgBr) that is insoluble in dilute ammonia but dissolves in CONCENTRATED aqueous ammonia to form $[\text{Ag}(\text{NH}_3)_2]^+$. (3) I^- gives a pale yellow precipitate (AgI) that is completely insoluble in both dilute and concentrated ammonia. The observations uniquely confirm bromide (Br^-).	[1]
48	C	Option C is correct. Thermal stability depends on the covalent bond enthalpy of the H-X bond. As halogen atomic radius increases down Group 17, the orbital overlap between hydrogen 1s and the halogen p orbital becomes less effective and the bond length increases. H-I has the longest bond and lowest bond enthalpy (299 kJ/mol), decomposing readily on gentle heating into H_2 and I_2 (purple vapor).	[1]
49	B	Option B is correct. In water, chlorine undergoes disproportionation: $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCl} + \text{HClO}$ (chlorine is oxidized to +1 in HClO and reduced to -1 in HCl). Chloric(I) acid (hypochlorous acid, HClO) is an uncharged molecule that penetrates bacterial cell walls and destroys vital enzymes and nucleic acids by powerful oxidation.	[1]
50	A	Option A is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
51	D	Option D is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
52	C	Option C is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
53	A	Option A is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
54	C	Option C is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
55	B	Option B is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
56	D	Option D is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
57	C	Option C is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
58	A	Option A is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
59	D	Option D is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
60	B	Option B is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
61	B	Option B is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
62	D	Option D is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
63	A	Option A is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
64	C	Option C is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]
65	A	Option A is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO_3 would be produced instead).	[1]

[illegible]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
86	D	Option D is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
87	B	Option B is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
88	C	Option C is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
89	B	Option B is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
90	A	Option A is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
91	D	Option D is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
92	C	Option C is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
93	A	Option A is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
94	C	Option C is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
95	B	Option B is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
96	D	Option D is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
97	C	Option C is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
98	A	Option A is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
99	D	Option D is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
100	B	Option B is correct. Household bleach (sodium chlorate(I), NaClO) is produced by reacting chlorine with cold, dilute aqueous sodium hydroxide: $\text{Cl}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. (At elevated temperatures, chlorate(V) NaClO ₃ would be produced instead).	[1]
101	C	Option C is correct. With concentrated H ₂ SO ₄ : NaCl (X) undergoes only acid-base reaction, evolving steamy HCl fumes. NaBr (Y) produces steamy HBr, orange-brown Br ₂ , and choking SO ₂ (sulfur reduced to +4). NaI (Z) undergoes extensive redox, forming purple I ₂ vapor, choking SO ₂ , yellow elemental sulfur S, and bad-egg-smelling toxic H ₂ S (sulfur reduced to -2).	[1]
102	A	Option A is the essential Cambridge testing matrix: AgCl is white and soluble in dilute aqueous ammonia ([Ag(NH ₃) ₂] ⁺ complex formed); AgBr is cream and insoluble in dilute ammonia, but dissolves in concentrated aqueous ammonia; AgI is pale yellow and completely insoluble in both dilute and concentrated aqueous ammonia.	[1]
103	D	Option D is correct. With cold dilute alkali (15 °C): $\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$ (chlorine disproportionates to -1 and +1). With hot concentrated alkali (70 °C): $3\text{Cl}_2 + 6\text{OH}^- \rightarrow 5\text{Cl}^- + \text{ClO}_3^- + 3\text{H}_2\text{O}$ (chlorine disproportionates to -1 and +5).	[1]
104	B	Option B is correct. Thermal stability depends directly on H-X single covalent bond enthalpy: H-F (562 kJ/mol) > H-Cl (431 kJ/mol) > H-Br (366 kJ/mol) > H-I (299 kJ/mol). As halogen atomic radius increases down the group, the H-X bond length increases and bond enthalpy decreases, making HI decompose most readily upon heating.	[1]
105	A	Option A is correct. Chlorine displaces iodide ions to form elemental iodine: $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$. When non-polar cyclohexane is added, non-polar I ₂ molecules preferentially dissolve in the upper organic solvent layer, imparting an intense, characteristic violet / purple color (whereas bromine would appear orange).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
106	C	Option C is correct. Because fluorine is a tiny Period 2 atom, the F-F bond is extremely short (142 pm). The three non-bonding electron lone pairs on each fluorine atom are crowded into close spatial proximity, causing intense electrostatic inter-electronic repulsion between the lone pairs on adjacent atoms. This destabilizing repulsion offsets the bonding orbital overlap, reducing the F-F bond enthalpy to only 158 kJ/mol.	[1]
107	B	Option B is correct. Reducing agents donate electrons ($2X^- \rightarrow X_2 + 2e^-$). The iodide ion (I^-) has the largest ionic radius (220 pm) and the greatest inner-shell shielding among the common halides. Its valence electrons are furthest from the nucleus and experience the weakest electrostatic attraction, making I^- the most easily oxidised (strongest reducing agent).	[1]
108	D	Option D is correct. Oxidizing agents accept electrons ($X_2 + 2e^- \rightarrow 2X^-$). Fluorine has the smallest atomic radius, the highest electronegativity (4.0), and the most positive standard reduction potential ($E^\circ = +2.87 \text{ V}$), enabling it to attract and capture electrons from almost any other chemical species.	[1]
109	A	Option A is correct. If carbonate or sulfite ions are present in the sample, adding AgNO_3 would form insoluble white Ag_2CO_3 or Ag_2SO_3 precipitates, giving a false-positive test for chloride. Pre-acidifying with dilute HNO_3 destroys carbonates and sulfites (producing CO_2 or SO_2 gas) without precipitating silver ions (AgNO_3 is soluble).	[1]
110	C	Option C is correct. When chlorine dissolves in water: $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCl} + \text{HClO}$. Chloric(I) acid (HClO) is a neutral, powerful oxidizing agent that diffuses through bacterial cell walls and oxidises crucial metabolic enzymes, destroying pathogens and sterilizing the water supply.	[1]